Digital Communication Analysis

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Communication Analysis (A11990)

Short Title:	Digital Communication Analysis
Faculty	Science and Computing
Department	Engineering Technology
Cluster	Electronics
Author	CLAIRE.FITZPATRICK
Credits	5
Level:	Advanced

Description of Module / Aims

Information theory and the restrictions and guidelines it presents for digital communication are treated. The role of probability in quantifying information and redundancy is treated leading to source coding, and channel coding. Signal space is presented and the error performance of M-ary FSK and PSK modulation schemes is developed. The bandwidth efficiency plane is then populated for these schemes, leading to power/bandwidth trade-offs and the Shannon limit. Satellite systems are explored as a case study in digital communications. Basic features of GPS, Radar and optical communications are surveyed.

Programmes

`COME-0003`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	4 / 7 / E
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	3 / 6 / M

Indicative Content

- Information Theory: data, information, entropy, equivocation, source coding, efficiency - Huffman, Shannon-Fano
- Channel coding: convolutional encoders, systematic/non-systemic, catastrophic error propagation.
- Modulation scheme bandwidth efficiency Performance
- Noise and Interference Sources: noise figure, noise temperature, low noise pre-amplifiers
- Satellite Systems: Frame structure, synchronisation, footprints, beamwidth/gain, link budget analysis.
- GPS & RADAR: Pseudo-ranges, code phase/carrier phase, ephemeris, almanac, GDOP, RADAR equation, ranging and curvature.
- Optical Systems: 1550nm & 1330nm characteristics, fibre events, Optical Time Domain Reflectometry

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Derive and calculate error rates for various modulation schemes and be able to devise a realistic choice of scheme for particular applications.
2. Discuss the origin and significance of various noise and interference sources and be able to compose a realistic link budget.
3. Explain source coding and be able to determine code solutions and associated parameters.
4. Outline basic channel coding approaches, and know key parameters & the relative strengths of approaches.
5. Describe the key parameters of GPS, optical comms & radar.
6. Write technical reports on selected problems using simulations as appropriate.

Learning and Teaching Methods

- Lectures
- Practicals

- Integrated Worked Examples

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3,4,5
Practical	20%	6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Practical	12	
Independent Learning	87	

Essential Material

- Sklar, B. _Digital Communications, Fundamentals and Applications_. 2nd ed.. New York: Prentice Hall, 2001.

Supplementary Material

- Glover, I.A. and P.M. Grant. _Digital Communications_. 3rd ed.. Harrow.: Pearson Education, 2010.
- Taub, H., D. Schilling and G. Saha. _Principles of Communication Systems_. 4th ed.. New York: McGraw Hill, 2008.

Requested Resources

- ENGINEERING LAB: Electronics